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AE6009 – Viscous Fluid Flow

Homework #2

Part A). Description of the equation and solution process

To solve the stagnation point flow problem, this program solves the following ODE:

$$f\frac{d^2f}{d\eta^2}+\frac{d^3f}{d\eta^3}+1-\left(\frac{df}{d\eta}\right)^2=0$$

By using an intermediate variable G where,

$$f\frac{dG}{d\eta}+\frac{d^2G}{d\eta^2}+1-G^2=0 \qquad G=\frac{df}{d\eta}$$

In practice, first, a grid of N points is initialized. The spacing between these points increases by 1.1x from 0 to N given by

$$\Delta\eta_j = K\Delta\eta_{j-1}$$

Next, an array of size N is initialized with our initial guess of G such that G(0) = 0, G(1→N) = 1

Since G = df/dn, we can solve for f array using the following equation:

$$f_{j+1} = f_j + \frac{G_j + G_{j+1}}{2}\Delta\eta_{j+1}$$

Now to approximate the ODE, the function must be approximated at N points in a system of linear equation using a central difference method.

$$\left.\frac{dG}{d\eta}\right|_j \approx \frac{G_{j+1}-G_{j-1}}{\Delta\eta_{j+1}+\Delta\eta_j}$$

The system of linear equations follow this formulation:

$$A_jG_{j-1}+B_jG_j+C_jG_{j+1}=-(\Delta\eta_j+\Delta\eta_{j+1})$$

Where,

$$A_j = -f_j + \frac{2}{\Delta\eta_j}$$

$$C_j = +f_j + \frac{2}{\Delta\eta_{j+1}}$$

$$B_j = -(A_j + C_j) - G_j(\Delta\eta_j + \Delta\eta_{j+1})$$

These values are assembled into a matrix to solve the equation MG = RHS:

$$\begin{bmatrix} 1 & & & & & & & \\ A_2 & B_2 & C_2 & & & & & \\ & A_3 & B_3 & C_3 & & & & \\ & & & A_j & B_j & C_j & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & A_{N-1} & B_{N-1} & C_{N-1} \\ 0 & & & & & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} G_1 \\ G_2 \\ G_3 \\ G_j \\ \\ \\ \\ G_{N-1} \\ G_N \end{Bmatrix} = \begin{Bmatrix} 0 \\ RHS_2 \\ RHS_3 \\ \\ \\ \\ \\ RHS_{n-1} \\ 1 \end{Bmatrix}$$

Where,

$$\mathrm{RHS_j} = -(\Delta \mathrm{n_j} + \Delta \mathrm{n_{j+1}})$$

Finally, solve for G using $\mathrm{G} = \mathrm{M}(\mathrm{RHS}^{-1})$. This gives an approximation for G, so to increase accuracy, it is important to iteratate this procedure by comparing the old G to the computed G to minimize error.

Part 2). Listing of the Computer Program

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%step 1: Selection of points between eta = 0 and eta > 10;

N = 40;
K = 1.1;
ETA = zeros(N, 1);
deltaETA = zeros(N, 1);
deltaETA(2) = 0.02;

for j = 3:N
   deltaETA(j) = K * deltaETA(j - 1);
end

for j = 2:N
   ETA(j) = ETA(j-1) + deltaETA(j);
end

%step 2: Initilizing the G vector
G = ones(40, 1);
G(1) = 0;

%iterating to converge G
error = Inf;
while error > 1e-3

   oldG = G;

   %step 3: Computing f from G
   f = zeros(N, 1);
   for j = 1:N-1
      f(j+1) = f(j) + (1/2) * (G(j) + G(j+1)) * deltaETA(j+1);
   end

   %step 4: discretization of G dif-e
   A = zeros(N-1, 1);
   B = zeros(N, 1);
   C = zeros(N-1, 1);

   for j = 2:N-1
      A(j-1) = -f(j) + 2 / (deltaETA(j));
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        C(j) =  f(j) + 2 / (deltaETA(j+1));
        B(j) = -(A(j-1) + C(j)) - G(j) * (deltaETA(j) + deltaETA(j+1));
    end

    %step 5: Solve the tridiagonal system Ax = b
    M = diag(B) + diag(A, -1) + diag(C, 1);
    M(1, :) = 0;
    M(1, 1) = 1;
    M(end, :) = 0;
    M(end, end) = 1;

    %step 6: apply boundary conditions
    RHS = zeros(N,1);
    for j = 2:N-1
        RHS(j) = - (deltaETA(j) + deltaETA(j+1));
    end
    RHS(N) = 1;

    %step 7: solve system of equations
    G = M \ RHS;

    error = max(abs(G - oldG));
end
```

Part C). Table comparing the output to known solution

η	*Expected f*	*Computed f*	*f Error (%)*	*Expected G*	*Computed G*	*G Error (%)*
0.0	0.000000	0.000000	N/A	0.000000	-6.6614e-16	N/A
0.1	0.005996	0.0060882	1.538	0.118265	0.11823	0.030
0.2	0.023322	0.023488	0.712	0.226612	0.22655	0.027
0.3	0.050995	0.051280	0.559	0.325241	0.32510	0.043
0.4	0.088057	0.088462	0.460	0.414456	0.41423	0.055
0.5	0.133585	0.133890	0.228	0.494649	0.49458	0.014
0.6	0.186701	0.187320	0.332	0.566281	0.56589	0.069
0.7	0.246573	0.247110	0.218	0.629861	0.62960	0.041
0.8	0.312423	0.312870	0.143	0.685937	0.68583	0.016
0.9	0.383529	0.384030	0.131	0.735079	0.73494	0.019
1.0	0.459227	0.459940	0.155	0.777865	0.77747	0.051
1.1	0.538910	0.539860	0.176	0.814873	0.81416	0.087
1.2	0.622028	0.623040	0.163	0.846671	0.84587	0.095
1.3	0.708089	0.708810	0.102	0.873808	0.87350	0.035
1.4	0.796652	0.797490	0.105	0.896809	0.89637	0.049
1.5	0.887329	0.888410	0.122	0.916168	0.91533	0.091
1.6	0.979779	0.980430	0.066	0.932348	0.93234	0.001
1.7	1.073710	1.074800	0.102	0.945774	0.94504	0.078
1.8	1.168860	1.169700	0.072	0.956834	0.95656	0.029
1.9	1.265010	1.266000	0.078	0.965877	0.96530	0.060
2.0	1.361970	1.362800	0.061	0.973217	0.97301	0.021
2.1	1.459600	1.460600	0.069	0.979128	0.97863	0.051
2.2	1.557760	1.558600	0.054	0.983853	0.98388	0.003
2.3	1.656340	1.657400	0.064	0.987601	0.98721	0.040
2.4	1.755250	1.756100	0.048	0.990549	0.99054	0.001
2.5	1.854430	1.855400	0.052	0.992851	0.99267	0.018
2.6	1.953810	1.954800	0.051	0.994634	0.99445	0.018
2.7	2.053340	2.054200	0.042	0.996003	0.99609	0.009
2.8	2.153000	2.154000	0.046	0.997046	0.99693	0.012
2.9	2.252740	2.253700	0.043	0.997834	0.99778	0.005
3.0	2.352560	2.353500	0.040	0.998424	0.99848	0.006
3.1	2.452420	2.453400	0.040	0.998863	0.99882	0.004
3.2	2.552330	2.553300	0.038	0.999186	0.99917	0.002
3.3	2.652260	2.653200	0.035	0.999422	0.99948	0.006
3.4	2.752210	2.753200	0.036	0.999593	0.99960	0.001
3.5	2.852170	2.853100	0.033	0.999716	0.99971	0.001
3.6	2.952150	2.953100	0.032	0.999803	0.99982	0.002
3.7	3.052130	3.053100	0.032	0.999865	0.99988	0.002
3.8	3.152120	3.153100	0.031	0.999908	0.99991	0.000
3.9	3.252110	3.253100	0.030	0.999938	0.99994	0.000

Part D). Plot comparing computed profile to exact solution

